GCE 2025 Chemistry Past Questions Analysis

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Section A: Multiple Choice Questions

Question 1

Question: An element X is found in period 2 and group III of the periodic table. X is not the usual symbol of the element. The valency of X is: A. 3 B. 6 C. 2 D. 5

Topic: Atomic Structure and Periodic Table **Concept:** Valency and Group Number **What is being tested:** The relationship between an element's group number and its valency. **Answer:** A. 3 **Why it is correct:** Elements in Group III have 3 electrons in their outermost shell. To achieve a stable octet, they typically lose these 3 electrons, giving them a valency of 3. **Why others are wrong:**

- B (6): This is not a common valency for Group III elements.
- C (2): This is the valency for Group II elements.
- D (5): This is the valency for Group V elements (if they gain 3 electrons) or their group number.

Question 2

Question: What is the symbol of the element with mass number 19 and atomic number 9? A. P B. F C. K D. Ne

Topic: Atomic Structure **Concept:** Atomic Number and Symbols **What is being tested:** Identification of an element using its atomic number. **Answer:** B. F **Why it is correct:** The atomic number (number of protons) uniquely identifies an element. The element with atomic number 9 is Fluorine (F). **Why others are wrong:**

- A (P): Phosphorus has atomic number 15.
- C (K): Potassium has atomic number 19.
- D (Ne): Neon has atomic number 10.

Question 3

Question: Which of the following reactions will lead to the liberation of hydrogen gas? A. Reaction of an acid with a base B. Reaction of an acid with a carbonate C. Reaction of an acid with a metal D. Reaction of an acid with a metal oxide

Topic: Acids, Bases, and Salts **Concept:** Reactions of Acids **What is being tested:** Knowledge of the products formed when acids react with different substances. **Answer:** C. Reaction of an acid with a metal **Why it is correct:** Acids react with reactive metals (above hydrogen in the reactivity series) to produce a salt and hydrogen gas. **Why others are wrong:**

- A: Acid + Base -> Salt + Water.
- B: Acid + Carbonate -> Salt + Water + Carbon Dioxide.
- D: Acid + Metal Oxide -> Salt + Water.

Question 4

Question: The charge of an electron relative to that of a proton is: A. equal and opposite B. the same C. negligible D. greater but opposite

Topic: Atomic Structure **Concept:** Subatomic Particles **What is being tested:** Comparison of the charges of subatomic particles. **Answer:** A. equal and opposite **Why it is correct:** An electron has a relative charge of -1, while a proton has a relative charge of +1. They are equal in magnitude but opposite in sign. **Why others are wrong:**

- B: They have opposite signs.
- C: The mass of an electron is negligible, not its charge.
- D: Their charges are equal in magnitude.

Question 5

Question: During the electrolysis of molten sodium chloride, using inert electrodes: A. A white solid is deposited at the anode B. Effervescence is seen at the cathode C. A pale green gas is liberated at the anode D. Hydrogen gas is discharged at the cathode

Topic: Electrochemistry **Concept:** Electrolysis of Molten Salts **What is being tested:** Predicting products of electrolysis for molten binary compounds. **Answer:** C. A pale green gas is liberated at the anode **Why it is correct:** In molten NaCl, Cl^- ions move to the anode (positive electrode) and are oxidized to chlorine gas (Cl_2), which is a pale green gas. **Why others are wrong:**

- A: Sodium metal (silvery) is deposited at the cathode, not a white solid at the anode.
- B: Sodium is deposited as a liquid/solid; no gas is produced at the cathode.
- D: There is no water in molten NaCl, so no hydrogen can be formed.

Question 6

Question: Calculate the percentage composition by mass of the water of crystallization in hydrated copper (II) sulphate pentahydrate ($CuSO_4.5H_2O$). RAM Cu = 64, S = 32, O = 16, H = 1. A. 7.2% B. 36% C. 10.1% D. 50.6%

Topic: Mole Concept and Chemical Calculations **Concept:** Percentage Composition **What is being tested:** Ability to calculate the mass percentage of a component in a compound. **Answer:** B. 36% **Why it is correct:**

- Molar mass of $CuSO_4.5H_2O = 64 + 32 + (4 \times 16) + 5 \times (2 \times 1 + 16) = 64 + 32 + 64 + 5(18) = 160 + 90 = 250$ g/mol.
- Mass of water = $5 \times 18 = 90$ g.
- % Water = $(90 / 250) \times 100 = 36\%$. **Why others are wrong:** These are incorrect calculations based on wrong molar masses or ratios.

Question 7

Question: Which of the following substances conduct electricity due to the presence of free mobile electrons? A. Aqueous sodium chloride B. Diamond C. Graphite D. Molten sodium chloride

Topic: Chemical Bonding / Periodicity **Concept:** Electrical Conductivity **What is being tested:** Distinguishing between ionic and metallic/covalent conduction. **Answer:** C. Graphite **Why it is correct:** Graphite has delocalized (free mobile) electrons between its layers, which allow it to conduct electricity. **Why others are wrong:**

- A & D: Conduct via mobile ions, not electrons.
- B: Diamond is an insulator as all its valence electrons are used in bonding.

Question 8

Question: Which of the following reactions is catalyzed by sunlight? A. Combustion of methane B. Decomposition of hydrogen peroxide C. Hydrolysis of starch D. Photosynthesis

Topic: Rates of Reaction / Organic Chemistry **Concept:** Photochemical Reactions **What is being tested:** Identifying reactions that require light energy. **Answer:** D. Photosynthesis **Why it is correct:** Photosynthesis is a photochemical process where plants use sunlight to convert CO_2 and water into glucose. **Why others are wrong:**

- A: Requires heat/ignition.
- B: Can be catalyzed by MnO_2 or light, but photosynthesis is the classic example of a light-driven reaction.
- C: Catalyzed by enzymes (amylase) or acids.

Question 9

Question: What is the property of concentrated sulphuric acid that makes it a good drying agent? A. It is cheap B. It is hygroscopic C. It is not deliquescent D. It is corrosive

Topic: Sulphur and its Compounds **Concept:** Properties of H_2SO_4 **What is being tested:** Understanding the drying action of concentrated sulphuric acid. **Answer:** B. It is hygroscopic **Why it is correct:** Hygroscopic substances absorb moisture from the air without necessarily dissolving in it, making them effective drying agents. **Why others are wrong:**

- A: Cost doesn’t determine chemical property.
- C: Deliquescence is a specific type of moisture absorption; being non-deliquescent doesn’t explain why it’s a good dryer.

- D: Corrosiveness is a safety hazard, not a drying mechanism.

Question 10

Question: A sample of an oxide contains 0.24g of Mg and 0.16g of oxygen. What is the empirical formula of the oxide? (Mg=24, O=16) A. MgO B. MgO_2 C. Mg_2O D. Mg_2O_3

Topic: Mole Concept **Concept:** Empirical Formula **What is being tested:** Calculating empirical formulas from mass data. **Answer:** A. MgO **Why it is correct:**

- Moles of Mg = 0.24 / 24 = 0.01 mol.
- Moles of O = 0.16 / 16 = 0.01 mol.
- Ratio Mg:O = 1:1. Formula = MgO. **Why others are wrong:** These ratios do not match the calculated mole ratio.

Question 11

Question: Which of the following metals will react vigorously with cold water producing Hydrogen gas? A. Beryllium B. Magnesium C. Iron D. Calcium

Topic: Metals / Reactivity Series **Concept:** Reaction of Metals with Water **What is being tested:** Knowledge of the reactivity of metals with water. **Answer:** D. Calcium **Why it is correct:** Calcium is high in the reactivity series and reacts steadily with cold water to produce hydrogen and calcium hydroxide. **Why others are wrong:**

- A: Beryllium does not react with water.
- B: Magnesium reacts very slowly with cold water (reacts with steam).
- C: Iron reacts only with steam when red hot.

Question 12

Question: Which of the following is a characteristic property of the transition elements? A. They exhibit only one oxidation state B. They generally have low densities C. They exhibit multiple oxidation states D. They are poor conductors of electricity

Topic: Periodic Table / Transition Metals **Concept:** Properties of Transition Metals **What is being tested:** Identifying unique features of transition elements. **Answer:** C. They exhibit multiple oxidation states **Why it is correct:** Transition metals have

partially filled d-orbitals, allowing them to lose different numbers of electrons and show variable valency. **Why others are wrong:**

- A: They show variable oxidation states.
- B: They are generally dense metals.
- D: They are excellent conductors.

Question 13

Question: Identify from the list below the element which is stored under paraffin oil. A. Calcium B. Sodium C. Phosphorus D. Aluminium

Topic: Metals / Group I Elements **Concept:** Storage of Reactive Metals **What is being tested:** Knowledge of the high reactivity of alkali metals. **Answer:** B. Sodium **Why it is correct:** Sodium is extremely reactive with oxygen and moisture in the air, so it must be stored under oil to prevent reaction. **Why others are wrong:**

- A: Calcium is reactive but usually stored in sealed containers.
- C: White phosphorus is stored under water.
- D: Aluminium is stable in air due to its oxide layer.

Question 14

Question: During a chemical reaction: A. bond breaking releases energy while bond formation requires energy B. both bond breaking and bond formation require energy C. both bond breaking and bond formation release energy D. bond breaking requires energy while bond formation releases energy

Topic: Energetics **Concept:** Bond Energies **What is being tested:** Understanding the energy changes during bond breaking and formation. **Answer:** D. bond breaking requires energy while bond formation releases energy **Why it is correct:** Breaking bonds is an endothermic process (requires energy input), while forming bonds is an exothermic process (releases energy). **Why others are wrong:** These options misstate the energy flow for these processes.

Question 15

Question: Which of the statements below is NOT true about an alloy? A. alloy is a mixture of two or more elements at least one of which is a metal B. An alloy has properties that are different from those of the individual elements C. An alloy is made up of two non-metals D. An alloy has improved properties than the individual elements

Topic: Metals / Alloys **Concept:** Definition and Properties of Alloys **What is being tested:** Identifying the defining characteristics of alloys. **Answer:** C. An alloy is made up of two non-metals **Why it is correct:** By definition, an alloy must contain at least one metal. A mixture of two non-metals is not an alloy. **Why others are wrong:** A, B, and D are all true statements about alloys.

Question 16

Question: A bottle of perfume opened in the front of a classroom can eventually be smelled at the back of the room. Select the word that best describes the behavior of the perfume. A. Diffusion B. Osmosis C. Filtration D. Conduction

Topic: Kinetic Theory of Matter **Concept:** Diffusion **What is being tested:** Identifying the process of gas particle movement. **Answer:** A. Diffusion **Why it is correct:** Diffusion is the net movement of particles from a region of higher concentration to a region of lower concentration. **Why others are wrong:**

- B: Osmosis involves water movement through a membrane.
- C: Filtration is a separation technique for solids and liquids.
- D: Conduction is heat or charge transfer.

Question 17

Question: When diluting concentrated sulphuric acid in the laboratory, the right procedure to follow is to: A. Add water to the acid while stirring B. Mix equal volumes of water and the acid C. Heat up the acid before adding water D. Add the acid to water slowly while stirring

Topic: Laboratory Safety / Acids **Concept:** Dilution of Concentrated Acids **What is being tested:** Knowledge of safe laboratory practices. **Answer:** D. Add the acid to water slowly while stirring **Why it is correct:** The dilution of H_2SO_4 is highly

exothermic. Adding acid to water allows the large volume of water to absorb the heat generated, preventing splashing. **Why others are wrong:**

- A: Adding water to acid can cause the acid to boil and splash out (dangerous).
- B: Doesn't specify the order of addition.
- C: Heating the acid makes it even more dangerous.

Question 18

Question: Which of the following contains a double covalent bond? A. Cl_2 B. O_2 C. N_2 D. H_2

Topic: Chemical Bonding **Concept:** Covalent Bonding **What is being tested:** Identifying the type of bond in diatomic molecules. **Answer:** B. O_2 **Why it is correct:** Oxygen atoms have 6 valence electrons and share two pairs to achieve an octet, forming a double bond (O=O). **Why others are wrong:**

- A & D: Single bonds.
- C: Triple bond.

Question 19

Question: The equation for the reaction taking place at the cathode during the electrolysis of aqueous copper (II) sulphate solution using graphite electrodes is: A. $2H^+ + 2e^- \rightarrow H_2$ B. $SO_4^{2-} + 2e^- \rightarrow SO_2$ C. $Cu^{2+} + 2e^- \rightarrow Cu$ D. $4OH^- \rightarrow 2H_2O + O_2 + 4e^-$

Topic: Electrochemistry **Concept:** Electrolysis of Aqueous Solutions **What is being tested:** Predicting cathode products in aqueous electrolysis. **Answer:** C. $Cu^{2+} + 2e^- \rightarrow Cu$ **Why it is correct:** Cu^{2+} ions are lower than H^+ in the electrochemical series, so they are preferentially discharged at the cathode to form copper metal. **Why others are wrong:**

- A: H^+ is not discharged because Cu^{2+} is present.
- B: This is not a standard electrode reaction.
- D: This is the anode reaction.

Question 20

Question: What is the product formed at the anode during the electrolysis of dilute sulphuric acid? A. Hydrogen gas B. Oxygen gas C. Copper metal D. Sulphur dioxide

Topic: Electrochemistry **Concept:** Electrolysis of Dilute Acids **What is being tested:** Predicting anode products. **Answer:** B. Oxygen gas **Why it is correct:** In dilute H_2SO_4, OH^- ions (from water) are discharged at the anode to produce oxygen gas. **Why others are wrong:**

- A: Hydrogen is formed at the cathode.
- C: No copper is present.
- D: Sulphate ions are not discharged in dilute solution.

Question 21

Question: Which of the following elements exists in nature as a diatomic molecule? A. carbon B. Oxygen C. phosphorus D. Magnesium

Topic: Chemical Bonding / Periodicity **Concept:** Diatomic Molecules **What is being tested:** Identifying elements that naturally form pairs of atoms. **Answer:** B. Oxygen **Why it is correct:** Oxygen exists as O_2 molecules. **Why others are wrong:**

- A: Carbon exists as a giant covalent structure (diamond/graphite).
- C: Phosphorus exists as P_4.
- D: Magnesium is a metal (monatomic lattice).

Question 22

Question: State what happens when calcium carbonate is treated with dilute hydrochloric acid. A. effervescence occurs and oxygen gas is released B. effervescence occurs and carbon dioxide gas is released C. white misty fumes are observed D. a white precipitate is formed

Topic: Acids, Bases, and Salts **Concept:** Reaction of Carbonates with Acids **What is being tested:** Identifying products of carbonate-acid reactions. **Answer:** B. effervescence occurs and carbon dioxide gas is released **Why it is correct:** Carbonates react with acids to produce salt, water, and carbon dioxide gas (which causes bubbling/effervescence). **Why others are wrong:**

- A: Oxygen is not produced.
- C: Misty fumes are characteristic of HCl gas, not this reaction.
- D: Calcium chloride is soluble, so no precipitate forms.

Question 23

Question: An exothermic reaction can be defined as: A. A reaction that releases heat energy B. A reaction that absorbs heat energy C. A reaction that occurs at constant temperature D. A reaction that occurs at constant pressure

Topic: Energetics **Concept:** Exothermic Reactions **What is being tested:** Definition of exothermic processes. **Answer:** A. A reaction that releases heat energy **Why it is correct:** “Exo” means out, “thermic” means heat. Exothermic reactions give out heat to the surroundings. **Why others are wrong:**

- B: This is endothermic.
- C & D: These are thermodynamic conditions, not definitions of heat flow.

Question 24

Question: What is the effect of carbon dioxide gas on a candle flame? A. It burns with a pop sound B. It causes the flame to burn brighter C. It puts out the flame D. It decreases the size of the flame

Topic: Non-metals / Carbon and its Compounds **Concept:** Properties of CO_2 **What is being tested:** Knowledge of the fire-extinguishing property of CO_2. **Answer:** C. It puts out the flame **Why it is correct:** Carbon dioxide does not support combustion and is denser than air, so it smothers the flame. **Why others are wrong:**

- A: This is the test for hydrogen.
- B: This is the effect of oxygen.
- D: It extinguishes it completely, not just reduces it.

Question 25

Question: Which of the following is (are) an example(s) of a giant atomic crystal? A. Diamond B. Ice C. Graphite D. Sodium Chloride

Topic: Chemical Bonding **Concept:** Giant Covalent Structures **What is being tested:** Distinguishing between different types of crystal lattices. **Answer:** A and C (Diamond and Graphite) **Why it is correct:** Both diamond and graphite consist of atoms linked by covalent bonds in a continuous 3D or 2D network. **Why others are wrong:**

- B: Ice is a molecular crystal.
- D: Sodium chloride is an ionic crystal.

Question 26

Question: Which of the following gas (es) is (are) responsible for global warming? A. Sulphur dioxide B. Carbon dioxide C. Hydrogen D. Methane

Topic: Environmental Chemistry **Concept:** Greenhouse Gases **What is being tested:** Identifying gases that contribute to the greenhouse effect. **Answer:** B and D (Carbon dioxide and Methane) **Why it is correct:** CO_2 and CH_4 are major greenhouse gases that trap heat in the atmosphere. **Why others are wrong:**

- A: Causes acid rain, not primarily global warming.
- C: Not a greenhouse gas.

Question 27

Question: The rate of the reaction $CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$ would NOT be influenced by: A. Temperature B. Surface area C. Concentration D. Pressure

Topic: Rates of Reaction **Concept:** Factors Affecting Rate **What is being tested:** Identifying factors that do not affect the rate of a specific reaction. **Answer:** D. Pressure **Why it is correct:** Pressure only significantly affects the rate of reactions involving gases as *reactants*. Here, the reactants are solid and aqueous. **Why others are wrong:** A, B, and C all affect the rate of this solid-liquid reaction.

Question 28

Question: By which of the following processes do plants remove carbon dioxide from the atmosphere? A. Respiration B. Photosynthesis C. Combustion D. Decomposition

Topic: Environmental Chemistry / Carbon Cycle **Concept:** Photosynthesis **What is being tested:** Understanding the carbon cycle. **Answer:** B. Photosynthesis **Why it is correct:** Plants take in CO_2 to make food during photosynthesis. **Why others are wrong:** A, C, and D all *release* CO_2 into the atmosphere.

Question 29

Question: Identify a property which is characteristic of halogens. A. They exist as solids, liquids and gases at room temperature B. They have high melting and boiling points C. They do not react with non-metals D. They are electrolytes

Topic: Periodic Table / Halogens **Concept:** Properties of Group VII **What is being tested:** Identifying physical and chemical trends in halogens. **Answer:** A. They exist as solids, liquids and gases at room temperature **Why it is correct:** Fluorine/Chlorine (gases), Bromine (liquid), Iodine (solid). **Why others are wrong:**

- B: They have low melting/boiling points.
- C: They react readily with non-metals (e.g., $H_2 + Cl_2$).
- D: They are non-metals and non-conductors.

Question 30

Question: What is the name of the species formed when an atom gains one or more electrons? A. cation B. radical C. anion D. molecule

Topic: Atomic Structure / Bonding **Concept:** Ion Formation **What is being tested:** Basic terminology of ions. **Answer:** C. anion **Why it is correct:** Gaining electrons (negative charge) results in a negatively charged ion called an anion. **Why others are wrong:**

- A: Formed by losing electrons.
- B: Species with unpaired electrons.
- D: Neutral group of atoms.

Question 31

Question: Which homologous series below reacts with carboxylic acids to produce esters? A. Alkenes B. Alcohols C. Alkanes D. Alkynes

Topic: Organic Chemistry **Concept:** Esterification **What is being tested:** Identifying the reactants for ester formation. **Answer:** B. Alcohols **Why it is correct:** Carboxylic acid + Alcohol -> Ester + Water (in the presence of an acid catalyst). **Why others are wrong:** Alkenes, alkanes, and alkynes do not undergo this specific reaction.

Question 32

Question: What kind of reaction takes place when methane is reacted with chlorine in the presence of sunlight? A. Substitution reaction B. Combustion reaction C. Elimination reaction D. Dehydration reaction

Topic: Organic Chemistry **Concept:** Reactions of Alkanes **What is being tested:** Identifying the type of reaction for alkanes. **Answer:** A. Substitution reaction **Why it is correct:** Alkanes undergo substitution where a hydrogen atom is replaced by a halogen atom. **Why others are wrong:**

- B: Reaction with oxygen.
- C & D: Not typical for methane.

Question 33

Question: One of the following is NOT a precaution taken when determining the heat of combustion of ethanol. A. Close the spirit lamp tightly to prevent evaporation of ethanol B. Stir the water constantly for even distribution of heat C. Use a wind shield to prevent the flame from being put out D. Take out the thermometer from time to time to properly read and record the temperature rise

Topic: Energetics / Organic Chemistry **Concept:** Experimental Calorimetry **What is being tested:** Knowledge of experimental errors and precautions. **Answer:** D. Take out the thermometer from time to time to properly read and record the temperature rise **Why it is correct:** Removing the thermometer would lead to heat loss and inaccurate readings. It should remain in the water. **Why others are wrong:** A, B, and C are all valid and necessary precautions.

Question 34

Question: A sulphite reacts with dilute hydrochloric acid to produce a gas. The gas produced is: A. Sulphur dioxide B. Chlorine gas C. Hydrogen gas D. Sulphur trioxide gas

Topic: Acids, Bases, and Salts **Concept:** Reaction of Sulphites **What is being tested:** Identifying the gas produced from sulphites. **Answer:** A. Sulphur dioxide **Why it is correct:** Sulphite + Acid -> Salt + Water + Sulphur dioxide. **Why others are wrong:** These gases are not produced in this reaction.

Question 35

Question: Which of the following oxides dissolve in water to produce a solution whose pH value is greater than 7? A. CO_2 B. NO_2 C. SO_2 D. Na_2O

Topic: Acids, Bases, and Salts / Periodicity **Concept:** Nature of Oxides **What is being tested:** Distinguishing between acidic and basic oxides. **Answer:** D. Na_2O **Why it is correct:** Sodium oxide is a basic oxide (metal oxide) that reacts with water to form sodium hydroxide, an alkaline solution (pH > 7). **Why others are wrong:** A, B, and C are non-metal oxides which form acidic solutions (pH < 7).

Question 36

Question: First statement: Sea water is a raw material for the manufacture of NaOH through electrolysis. Second statement: Chlorine and hydrogen gases are important by-products of this process.

Topic: Electrochemistry / Industrial Chemistry **Concept:** Electrolysis of Brine **What is being tested:** Understanding the industrial process for manufacturing sodium hydroxide and its by-products. **Answer:** A. Both statements are true and the second statement is the correct explanation of the first. **Why it is correct:** Sea water (brine) is indeed the raw material for the Chlor-alkali process. The electrolysis of concentrated NaCl(aq) produces NaOH at the cathode, while Cl_2 and H_2 are liberated as valuable by-products. **Why others are wrong:** B, C, and D are incorrect because both statements are factually true and directly related to the same industrial process.

Question 37

Question: First statement: The ammonia molecule is trigonal pyramidal in shape. Second statement: The ammonia molecule has three bonded pairs and one lone pair of electrons.

Topic: Chemical Bonding **Concept:** VSEPR Theory / Molecular Shapes **What is being tested:** Relationship between electron pair geometry and molecular shape. **Answer:**

A. Both statements are true and the second statement is the correct explanation of the first. **Why it is correct:** Ammonia (NH_3) has 4 electron pairs (3 bonding, 1 lone). According to VSEPR theory, the lone pair repels the bonding pairs more strongly, resulting in a trigonal pyramidal shape instead of a perfect tetrahedron. **Why others are wrong:** The second statement is the direct reason for the shape described in the first statement.

Question 38

Question: First statement: Phosphorus is used in the manufacture of matches. Second statement: The allotropes of phosphorus are rhombic and monoclinic phosphorus.

Topic: Non-metals / Phosphorus **Concept:** Allotropy and Uses **What is being tested:** Knowledge of phosphorus allotropes and their applications. **Answer:** C. First statement is true, second statement is false. **Why it is correct:** Red phosphorus is indeed used in matches. However, rhombic and monoclinic are allotropes of **Sulphur**, not phosphorus. The common allotropes of phosphorus are white and red phosphorus. **Why others are wrong:** The second statement is factually incorrect regarding phosphorus.

Question 39

Question: Select from the methods below, a method which can be used to prepare silver chloride in the laboratory. A. Distillation of a mixture of salts B. Titration between an acid and a base C. Double Decomposition reaction D. Reaction between a metal and an acid

Topic: Acids, Bases, and Salts **Concept:** Preparation of Salts **What is being tested:** Knowledge of solubility rules and salt preparation methods. **Answer:** C. Double Decomposition reaction **Why it is correct:** Silver chloride (AgCl) is an insoluble salt. Insoluble salts are prepared by precipitation (double decomposition) by mixing two soluble salt solutions (e.g., $AgNO_3$ and NaCl). **Why others are wrong:**

- A: Distillation is for separating liquids.
- B: Titration is for soluble salts.
- D: Silver is below hydrogen in the reactivity series and won't react with dilute acids.

Question 40

Question: Why is lime water added to acidic soils? A. To lower the pH of the soil B. To improve the soil's drainage C. To neutralize excess acidity D. To increase the soil's capacity to hold water

Topic: Acids, Bases, and Salts **Concept:** Neutralization in Agriculture **What is being tested:** Practical application of neutralization reactions. **Answer:** C. To neutralize excess acidity **Why it is correct:** Lime (calcium hydroxide/oxide) is basic. It reacts with the acids in the soil to neutralize them, raising the pH to a level suitable for plant growth. **Why others are wrong:**

- A: It raises the pH (makes it less acidic).
- B & D: These are physical properties not directly addressed by the chemical addition of lime.

Question 41

Question: Fertilizer rating can be defined as: A. A measure of the acidity of the fertilizer B. A measure of the nutrient content of the fertilizer C. A measure of the water-holding capacity of the fertilizer D. A measure of the basicity of the fertilizer

Topic: Industrial Chemistry / Fertilizers **Concept:** Fertilizer Composition **What is being tested:** Definition of fertilizer rating (NPK). **Answer:** B. A measure of the nutrient content of the fertilizer **Why it is correct:** Fertilizer rating (e.g., 10-10-10) indicates the percentage by mass of the primary nutrients: Nitrogen (N), Phosphorus (P), and Potassium (K). **Why others are wrong:** Acidity, basicity, and water-holding capacity are not what the "rating" refers to.

Question 42

Question: Which of the following is an example of a condensation polymer? A. Polystyrene B. Polyethene C. nylon D. perspex

Topic: Organic Chemistry **Concept:** Polymerization **What is being tested:** Distinguishing between addition and condensation polymers. **Answer:** C. nylon **Why it is correct:** Nylon is formed by the reaction of monomers with the elimination of a small molecule (usually water), which is the definition of condensation

polymerization. **Why others are wrong:** A, B, and D are all addition polymers formed by the linking of unsaturated monomers without loss of atoms.

Question 43

Question: Which of the following is an example of sublimation in everyday life? A. Ice turning to liquid water in the sun B. Water drying out from a pot on the fire C. Rainfall D. Camphor balls disappearing when left in a box

Topic: Kinetic Theory of Matter **Concept:** Changes of State **What is being tested:** Identifying sublimation (solid to gas). **Answer:** D. Camphor balls disappearing when left in a box **Why it is correct:** Camphor (naphthalene) changes directly from a solid to a gas at room temperature without passing through the liquid state. **Why others are wrong:**

- A: Melting (solid to liquid).
- B: Evaporation (liquid to gas).
- C: Condensation (gas to liquid).

Question 44

Question: Calculate the number of particles found in 0.5 moles of a substance. A. 6.023×10^{23} B. 1.511×10^{23} C. 3.011×10^{23} D. 3.022×10^{24}

Topic: Mole Concept **Concept:** Avogadro's Constant **What is being tested:** Relationship between moles and number of particles. **Answer:** C. 3.011×10^{23} **Why it is correct:** Number of particles = Moles $\times$ Avogadro's number = $0.5 \times 6.022 \times 10^{23} = 3.011 \times 10^{23}$. **Why others are wrong:** These are incorrect multiples or divisions of Avogadro's constant.

Question 45

Question: In an acid/base titration, the role of the indicator is to: A. signal the end point of the titration B. determine the concentration of the base C. determine the pH of the solution D. serve as a catalyst

Topic: Acids, Bases, and Salts **Concept:** Titration **What is being tested:** Function of indicators in volumetric analysis. **Answer:** A. signal the end point of the titration **Why**

it is correct: Indicators change color at a specific pH range, which signals that the neutralization is complete (the end point). **Why others are wrong:**

- B: Calculations determine concentration.
- C: pH meters or universal indicators are used for pH measurement.
- D: Indicators do not speed up the reaction.

Question 46

Question: What is the role of atmospheric oxygen in the formation of ethanoic acid from ethanol? A. It acts as a catalyst B. It acts as an oxidizing agent C. It acts as a reducing agent D. It increases the pH of the solution

Topic: Organic Chemistry **Concept:** Oxidation of Alcohols **What is being tested:** Understanding the chemical role of oxygen in oxidation. **Answer:** B. It acts as an oxidizing agent **Why it is correct:** The conversion of ethanol (CH_3CH_2OH) to ethanoic acid (CH_3COOH) involves the addition of oxygen/removal of hydrogen, which is oxidation. Oxygen is the agent that causes this. **Why others are wrong:** Oxygen is consumed in the reaction (not a catalyst) and gains electrons/hydrogen (it is reduced, so it is an oxidizing agent).

Question 47

Question: Identify the reagent used to detect the presence of chloride ions in a compound. A. $BaCl_2$/HCl B. HNO_3 /$AgNO_3$/NH_3 C. NaOH D. CONC H_2SO_4/$FeSO_4$

Topic: Qualitative Analysis **Concept:** Identification of Anions **What is being tested:** Standard test for chloride ions. **Answer:** B. HNO_3 /$AgNO_3$/NH_3 **Why it is correct:** Adding silver nitrate ($AgNO_3$) acidified with nitric acid (HNO_3) to a chloride solution produces a white precipitate of AgCl, which dissolves in aqueous ammonia (NH_3). **Why others are wrong:**

- A: Test for sulphate ions.
- C: Test for cations.
- D: Brown ring test for nitrate ions.

Question 48

Question: What is the main cause of hardness in water? A. presence of sulphate ions B. presence of chloride ions C. presence of nitrate ions D. presence of calcium and magnesium ions

Topic: Water **Concept:** Hardness of Water **What is being tested:** Identifying the ions responsible for water hardness. **Answer:** D. presence of calcium and magnesium ions **Why it is correct:** Hardness is caused by dissolved polyvalent metallic ions, primarily Ca^{2+} and Mg^{2+}, which react with soap to form scum. **Why others are wrong:** While these ions might be present as counter-ions (e.g., $CaSO_4$), it is the metal cations themselves that cause the hardness.

Question 49

Question: Identify the use of carbon in the following reaction: $C + CO_2 \rightarrow 2CO$ A. an oxidizing agent B. a catalyst C. a reducing agent D. an inhibitor

Topic: Redox Reactions / Carbon **Concept:** Reducing Agents **What is being tested:** Identifying the role of a substance in a redox reaction. **Answer:** C. a reducing agent **Why it is correct:** Carbon © removes oxygen from CO_2 to form CO. Since it removes oxygen, it is a reducing agent (and is itself oxidized to CO). **Why others are wrong:** It is a reactant, not a catalyst or inhibitor, and it causes reduction, not oxidation of the other species.

Question 50

Question: Which of the following is an example of a strong electrolyte? A. Distilled water B. Sugar solution C. Vinegar D. Salt solution

Topic: Electrochemistry **Concept:** Electrolytes **What is being tested:** Distinguishing between strong, weak, and non-electrolytes. **Answer:** D. Salt solution **Why it is correct:** A salt solution (like NaCl in water) dissociates completely into ions, allowing it to conduct electricity very well. **Why others are wrong:**

- A: Very poor conductor (non-electrolyte).
- B: Non-electrolyte (molecules don’t ionize).
- C: Weak electrolyte (partially ionizes).

Section B: Structured Questions

Question 1 (from Page 2)

Question: The table below gives information about the atom X and the ions Y^+ and Z^{2+}.

Atoms/ions	Mass number	Protons	Electrons	Neutrons
X	23	A	11	B
Y^+	14	C	D	7
Z^{2+}	E	12	F	12

a. Complete the table by giving the values of A, B, C, D, E and F. b. i) Identify the Cation and Anion. ii) Write down the formula of the compound formed between Z^{2+} and Y^{3-}. iii) Identify the bond type that exists in b(ii) above.

Topic: Atomic Structure and Bonding **Concept:** Subatomic Particles and Ionic Bonding **Analysis:**

- For X (neutral atom): Protons = Electrons = 11. So **A = 11**. Neutrons = Mass - Protons = 23 - 11 = 12. So **B = 12**.
- For Y^+ (ion with +1 charge): Protons = Mass - Neutrons = 14 - 7 = 7. So **C = 7**. Electrons = Protons - Charge = 7 - 1 = 6. So **D = 6**.
- For Z^{2+} (ion with +2 charge): Mass = Protons + Neutrons = 12 + 12 = 24. So **E = 24**. Electrons = Protons - Charge = 12 - 2 = 10. So **F = 10**.

Answers: a. A=11, B=12, C=7, D=6, E=24, F=10. b. i) Cation: Y^+ and Z^{2+} (positively charged). Anion: None in the table (but Y^{3-} is mentioned later). ii) Formula: Z_3Y_2 (Z has 2+, Y has 3-). iii) Bond type: Ionic bond (between metal ion and non-metal ion).

Question 2 (from Page 2)

Question: The elements chlorine (Cl), Bromine (Br), Iodine (I) and Fluorine (F) belong to the same group of the periodic table. a. What is the family name given to this group of elements? b. Using their symbol only, arrange these elements in order of increasing reactivity. c. Identify a member of this group that exists as a liquid at room temperature. d. i. Define a diatomic molecule. ii. Write down the formula of Fluorine

molecule. iii. Apart from these elements name any other element that exists as a diatomic molecule. e. Complete and balance the following equations: (i) KBr + Cl_2 -> (ii) NaCl + Br_2 -> f. One of these elements is used in the treatment of tap water, identity this element.

Topic: Group VII Elements (Halogens) **Concept:** Periodicity and Reactivity **Answers:** a. Halogens. b. I < Br < Cl < F (Reactivity increases up the group). c. Bromine (Br). d. i. A molecule consisting of two atoms of the same or different elements chemically bonded together. ii. F_2. iii. Hydrogen (H_2), Nitrogen (N_2), or Oxygen (O_2). e. (i) 2KBr + Cl_2 -> 2KCl + Br_2 (ii) NaCl + Br_2 -> No reaction (Bromine is less reactive than Chlorine). f. Chlorine.

Question 3 (from Page 2)

Question: a. Sketch an energy level diagram representing an endothermic reaction. b. An ethanol lamp containing some ethanol weighed 35.56g. When the ethanol burnt for some time, it was found that the new mass of the lamp and ethanol was 34.56g and 29.7 $KJmol^{-1}$ of energy was evolved. i. Define heat of combustion. ii. What mass of ethanol was burnt? iii. Calculate the heat of combustion of ethanol of molar mass 46g.

Topic: Energetics **Concept:** Enthalpy Changes and Calculations **Answers:** a. (Diagram description: Reactants at lower energy level, Products at higher energy level, ΔH is positive). b. i. The heat energy released when one mole of a substance is completely burnt in excess oxygen. ii. Mass burnt = 35.56g - 34.56g = 1.00g. iii. 1.00g evolved 29.7 kJ. Moles of ethanol = 1.00 / 46 = 0.0217 mol. Heat of combustion = 29.7 / 0.0217 = 1368.2 kJ/mol.

Section C: Long Answer Questions

Question 8

Question: Atoms of elements are linked by chemical bonds which could be ionic, simple covalent or metallic. a. Define a chemical bond. b. Using suitable examples and diagrams describe how each bond is formed. c. State one property of each substance in (b) and relate the property to the bond type.

Topic: Chemical Bonding **Concept:** Types of Bonding **Answers:** a. A chemical bond is a lasting attraction between atoms, ions or molecules that enables the formation of chemical compounds. b.

- **Ionic:** Transfer of electrons from a metal to a non-metal (e.g., NaCl).
- **Covalent:** Sharing of electron pairs between non-metals (e.g., H_2O).
- **Metallic:** Attraction between positive metal ions and a sea of delocalized electrons (e.g., Cu). c.
- **Ionic (NaCl):** High melting point due to strong electrostatic forces in the lattice.
- **Covalent (H_2O):** Low boiling point due to weak intermolecular forces.
- **Metallic (Cu):** Good electrical conductivity due to mobile delocalized electrons.

Question 9

Question: Identify the gases described below. Briefly describe how a pure dry sample of each gas can be prepared and collected in the laboratory. i. It turns acidified potassium dichromate from orange to green without a precipitate. ii. It is odourless, and rekindles a glowing splint. iii. It burns with a 'pop' sound.

Topic: Identification of Gases **Concept:** Laboratory Preparation of Gases **Answers:** i. **Sulphur dioxide (SO_2):** Prepared by reacting sodium sulphite with dilute HCl. Dried by passing through concentrated H_2SO_4. Collected by downward delivery. ii. **Oxygen (O_2):** Prepared by catalytic decomposition of H_2O_2 using MnO_2. Dried by passing through anhydrous $CaCl_2$. Collected over water or by upward delivery. iii. **Hydrogen (H_2):** Prepared by reacting Zinc with dilute HCl. Dried by passing through anhydrous $CaCl_2$. Collected by upward delivery (less dense than air).

Question 10

Question: You are given the following conversions: a. CH_3CH_2OH -> C_2H_4 b. CH_3CH_2 -> $(CH_2CH_2CH_2CH_2-)n$ c. Fe -> Fe^{3+} d. Br_2 -> Br^- Describe how each of these conversions is carried out stating clearly the reagents used and corresponding reactions/equations.

Topic: Organic and Inorganic Conversions **Concept:** Chemical Reactions **Answers:** a. **Dehydration of Ethanol:** Heat ethanol with concentrated H_2SO_4 at 170°C. Equation: CH_3CH_2OH -> C_2H_4 + H_2O. b. **Polymerization of Ethene:** High pressure and temperature with a catalyst. Equation: n(C_2H_4) -> (-CH_2-CH_2-)n. c. **Oxidation of Iron:**

React iron with chlorine gas. Equation: $2Fe + 3Cl_2 \rightarrow 2FeCl_3$. d. **Reduction of Bromine:** React bromine with a reducing agent like potassium iodide. Equation: $Br_2 + 2KI \rightarrow 2KBr + I_2$.